



GENERATIVE AI

Assignment 2

Project Setup Guide

Group 12

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1 Project Setup Manual

This section describes the minimal steps required to clone, configure, and run the project locally.

1.1 Prerequisites

Before starting, ensure the following software is installed:

- Python 3.8 or higher
- Node.js 16 or higher
- npm package manager
- Git

A Firebase project with Authentication and Storage enabled is also required.

1.2 Repository Clone

Clone the repository from GitHub:

```
git clone https://github.com/FranciscoLemos03/GenAI-PR-2025w.git
cd ai-with-frontend
```

1.3 Backend Setup

1. Navigate to the backend directory:

```
cd api
```

2. Create and activate a virtual environment:

```
python -m venv venv

# Windows:
venv\Scripts\activate

# macOS/Linux:
source venv/bin/activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Start the backend server:

```
$env:PYTHONPATH=".."
python -m uvicorn app.main:app --reload
```

The API will be available at: (Note: it is important to ensure that the API is running on port 8000.)

```
http://localhost:8000
```

1.4 AI Setup

1. Navigate to the backend directory:

```
cd genai
```

2. Create and activate a virtual environment:

```
python -m venv venv

# Windows:
venv\Scripts\activate

# macOS/Linux:
source venv/bin/activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Add .env file with your GEMINI API Key

```
GEMMA_API_KEY=YOUR_KEY
```

1.5 Frontend Setup

1. Open a new terminal and navigate to the frontend folder:

```
cd frontend
```

2. Install dependencies:

```
npm install
```

3. Create a `.env` file and add the Firebase configuration:

```
REACT_APP_API_KEY=your_firebase_api_key  
REACT_APP_AUTH_DOMAIN=your_project.firebaseio.com  
REACT_APP_PROJECT_ID=your_firebase_project_id  
REACT_APP_STORAGE_BUCKET=your_firebase_project.appspot.com  
REACT_APP_MESSAGING_SENDER_ID=your_firebase_sender_id  
REACT_APP_APP_ID=your_firebase_app_id
```

4. Start the frontend server:

```
npm start
```

The web interface will be available at: (Note: it is important to ensure that the frontend is running on port 3000.)

```
http://localhost:3000
```

1.6 Running the System

To run the full system, both servers must be active:

- Backend running on port 8000
- Frontend running on port 3000

Once running, open a browser and access the frontend URL. Users can authenticate, create rooms, upload PDF documents, and perform semantic searches through the interface.